

TrES-3b

R-0622

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_260620 · created 2026-07-25T12:51:30+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461211.8838 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.015
Transit depth (Rp/Rs)^2	3.86 +/- 0.58 [%]
Orbital Inclination (inc)	81.98 +/- 0.47
Airmass coefficient 1 (a1)	0.99 +/- 0.013
Airmass coefficient 2 (a2)	0.009 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	#2 - [470, 436]
Optimal Aperture	3.82
Optimal Annulus	11.01
Transit Duration (day)	0.0598 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.015 vs 0.16309 (deviation 2.26σ)
Duration fit vs published	0.0598 d vs 0.0567 d (deviation 0.75σ)
Mid-time O–C	-5.7 min
Depth SNR	13.1
Residual scatter	1.3 %

Light curve

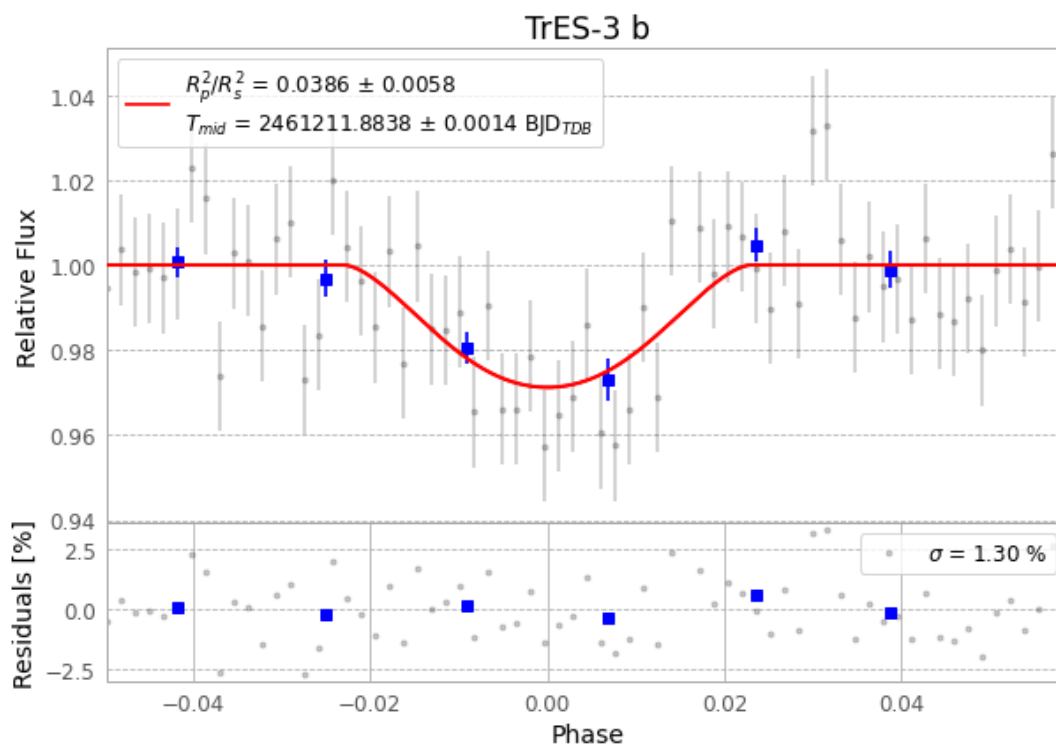


Figure 1 — FinalLightCurve_TrES-3 b_2026-06-20.png (SHA-256 61f336c056653d71...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [225, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e4d0383c26ef6ef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

69 FITS frames (45 MB), TRES-3260620073317.FITS → TRES-3260620105715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260620.log (32329109cd3e...), FinalLightCurve_TrES-3 b_2026-06-20.png (61f336c05665...), FinalLightCurve_TrES-3 b_2026-06-20.csv (31e01dc85964...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 12:51Z.